ABHINAV KAUSHIK

Senior Applied Scientist

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SUMMARY

Senior Applied Scientist with 9+ years in ML, NLP and LLMs who works backwards from customer problems and owns multi-agent and RAG systems end to end, from research to production, across Healthcare, Telecom and AIOps. Dives deep and delivers measurable results: 98% less manual analysis, 60% lower token cost, 43% less human review. Leads by developing people and raising the bar: has led teams of 3–4 engineers, mentored a team through 15+ research prototypes with 7 shipped to production or paid work, and set the evaluation standards and 11 reusable agent skills junior engineers build on. Also building ThinkTree.AI, a non-profit AI learning platform used by NGO teachers to support students with ADHD.

PROFESSIONAL EXPERIENCE

Senior Applied Scientist | United Health Group — Bengaluru, India | Nov 2025 - Present

- Led 3 engineers building and deploying a production lab-extraction agent with Claude Opus 4.7 as frontier model, Textract on an ECS cluster, Pydantic-validated outputs across 1,800 documents and work behind 800 Jira tickets.
- Engineered a production multi-agent analytics chatbot in Python with Pydantic-validated tool inputs, answering AARP medical-plan questions with PySpark analysis and multi-turn reasoning, cutting manual analysis by 98%.
- Architected a 6-agent RAG platform on AWS Bedrock for policy Q&A (design phase): HyDE query rewriting, hybrid BM25 and Titan retrieval, knowledge-graph extraction, grounded generation and LLM-as-judge hallucination checks.
- Built an automatic UI generator for customer delivery that turns Jira stories and requirements into 36+ discipline-specific UI mockups using HTML, Next.js and Claude 4.6, speeding up solution design.
- Standardized reproducible LLM evaluation, traceability and observability for the production analytics chatbot and its QA and Summary Agents, with LLM-judge metrics, tracing and failure detection, cutting human review by 43%.
- Owned token economics for agentic workflows, cutting cost up to 60% by benchmarking model choice per step, routing extraction and classification to smaller models and keeping frontier models for multi-step reasoning.
- Fine-tuned a cross-encoder on positive/negative pairs to re-rank RAG retrieval over AARP surveys, with embeddings served from Databricks Vector Search and pgvector, lifting question-answering accuracy over the baseline.

Senior Machine Learning Engineer | Viavi Solutions — Chennai, India | Sep 2024 - Nov 2025

- Led the team of 4 to build an event-correlation engine clustering alarms with Sentence-BERT and DBSCAN, tracing network alarm floods back to the root event and improving semantic alarm classification by 83%.
- Wrote 11 reusable LLM agent skills for the junior team, automating SDLC security-breach checks and test-case generation on post-development code and making post-release review repeatable.
- Deployed and owned a production SLA breach prediction system with LightGBM, FastAPI and MLflow, using GPT-3.5 for real-time root-cause analysis and remediation per breach, reducing downtime by 67%.
- Earned sales and presales trust with an internal RAG-based Project Discovery Assistant in typed Python: semantic retrieval over existing solutions grounded every LLM answer, with PII redaction, reaching a 90% match rate.
- Established CI/CD and experiment-tracking standards for model deployment with Jenkins, Bitbucket and Docker, adding unit and integration tests so every model release was repeatable and reviewed before production.
- Led and mentored a team of 4 engineers across 15+ AI research prototypes for AIOps and GeoPerformance clients in NAM, EMEA and LATAM, with 7 validated into production or paid client work.

Data Scientist | Viavi Solutions — Bengaluru, India | Nov 2021 - Sep 2024

- Shipped anomaly detection and demand forecasting on 15+ KPIs across 500+ telecom network devices using Isolation Forest and Holt-Winters, validated against historical baselines for proactive network planning.
- Contributed to network-device inventory demand forecasting using predictive models to improve planning and reduce revenue-at-risk; targeted a conservative 10% reduction in excess/at-risk inventory and higher ARPU.
- Trained and deployed a production time-series forecasting framework combining LSTM and XGBoost to model call quality, SMS delivery rates and network efficiency metrics for the Delta cellular network.
- Engineered a unified time-series pipeline in SQL and Python to ingest call logs, SMS delivery statistics and signal metrics across time intervals, joined by user ID and network location for downstream forecasting.
- Designed a geo-location dashboard mapping faulty network connections across EMEA, used by service providers to resolve faults at 950+ sites a year and cut fault turnaround time (TAT) by 77%.

Data Analyst | Optum — Gurugram, India | Jun 2021 - Nov 2021

- Built a customer-centric analytics engine and dashboard for the Renew Rewards programme for premium clients, applying Python NLP to member feedback to surface themes and sentiment, helping the marketing team convert 83% of targeted clients month over month.
- Automated Renew Rewards data processing and reporting in Python on Databricks, handling highly volatile member data in repeatable pipelines that kept analytics and marketing teams working from current numbers.

Data Engineer | Nippon Data Systems — Gurugram, India | Nov 2019 - May 2021

- Engineered SQL, PySpark and Airflow pipelines automating BAU medical-audit reporting on large healthcare datasets, cutting manual reporting effort by 95%.

- Wrote complex SQL queries with multi-table joins, CTEs and window functions, delivering ad-hoc reports and analyses for multiple stakeholder teams.
- Built real-time log observability for the COVID QRT team with Elasticsearch and Kibana, improving traceability efficiency by 80%.

Associate Consultant | Saksoft Gurugram, India | Dec 2017 - Nov 2019

- Built customer dashboards for financial and product insights using QlikSense, SQL and SAP HANA, increasing user engagement by 40%.
- Developed an XGBoost propensity model predicting personal-loan applications within 7 days of contact, enabling targeted outreach.
- Wrote complex SAP HANA SQL with CTEs, joins, subqueries, window functions and aggregations to extract and validate financial modelling data.
- Built reusable SQL data-preparation workflows across financial and product datasets, refining queries and filters to surface relevant signals for financial teams.

PROJECTS

Think Tree | ML Study Platform

- Built a non-profit AI-powered ML study platform for people with ADHD and context-switching challenges, designed to help users organize learning and maintain study context.

Grounded Customer Support Agent

- Built an agentic customer-support assistant for online orders and returns, combining PydanticAI tool calling, RAG with a PyTorch cross-encoder re-ranker, and a grounding check against hallucinated answers.
- Evaluated it with a reproducible harness of labelled support conversations and LLM judges for faithfulness, tracking runs in MLflow on Databricks and serving it through FastAPI with tracing.
- **Technologies:** PydanticAI, PyTorch, Databricks Vector Search, RAG, LLM-as-Judge, MLflow, FastAPI, Python

E-Commerce Post-Deployment Security Agent

- Built an agentic security tool that evaluates live e-commerce sites post-deployment, driving a real browser against a 144-control rule pack with Playwright and an LLM to interpret evidence and classify threats.
- Validated across 20+ e-commerce websites from large-scale platforms to recently vibe-coded storefronts, surfacing post-deployment security threats with structured reports, automated screenshots and reproducible evidence.

Auto Evals Bot | Live system

- Built and run a live Python bot that writes Pydantic-typed evaluation cases for LLM and agent features on demand, turning sample inputs and expected behaviour into regression suites with traced, observable runs.

RFP Multi-Agent Proposal Automation Platform | Live system

- Built and run a live multi-agent RFP platform in Python with Pydantic schemas: an LLM supervisor orchestrates intent detection, retrieval, proposal generation and document matching, with RAG over past proposals and RBAC.

EDUCATION

Master of Science, Machine Learning and AI | Liverpool John Moores University Jan 2021 - Jan 2023
Bachelor of Technology, Information Technology | Jamia Hamdard Aug 2013 - Aug 2017

TECHNICAL SKILLS

AI / GenAI: Generative AI, LLM, RAG, Agentic RAG, Agentic AI, LangGraph, LangChain, Claude, OpenAI, Embeddings, Cross-Encoder, Fine-Tuning, Hugging Face Transformers, NLP, Tool Orchestration, Amazon Textract
Machine Learning: PyTorch, Scikit-learn, XGBoost, LightGBM, Random Forest, Isolation Forest, LSTM, SBERT, DBSCAN, Time Series Forecasting, Demand Forecasting, Anomaly Detection, Statistical Modeling, Regression, Deep Learning, A/B Testing
Evals / Observability: LLM Evals, LLM Evaluators, Agent Tracing, Observability, Elasticsearch, Kibana, Log Analysis
Programming / Data: Python, SQL, PySpark, Pandas, NumPy, FastAPI, Flask, Hive, Spark SQL, REST API, Pydantic, Unix
MLOps / Cloud: MLflow, Docker, Kubernetes, Airflow, DVC, Terraform, Jenkins, Bamboo, Bitbucket, CI/CD, GCP, Azure, Vercel
Data Platforms: Databricks, Databricks SQL, Databricks Vector Search, BigQuery, Apache Spark, Supabase
Databases: TimescaleDB, MySQL, Vector DB, pgvector, ArangoDB, Cosmos DB
Search: Apache Solr, Lucene Indexing, Information Retrieval
BI: Power BI, QlikSense, SAP HANA

LANGUAGES

English: Native
German: A1, Pursuing

ADDITIONAL INFORMATION

- Network Intern, Airport Authority of India

- Hadoop Intern, Ducat Technologies